Course Summary: DAIDP

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Understanding Your Audience

Psychographics

Psychographics analyzes consumers based on their activities, interests, and opinions - going beyond demographics.

Personas

A fictional user profile for a product or service.

Customer Journey

A customer journey maps the full experience of interacting with a product or brand.

Understanding audience is the essential first step in creating effective, human-centered AI products that truly resonate with users, and deliver meaningful value throughout the user experience.

Human-Machine Interaction

Master-Servant

Hierarchical dynamic, the human takes the dominant role—issuing commands, while the machine executes specific, predefined tasks.

Example - Amazon Alexa

Partner-Partner

Interaction becomes a dynamic exchange. This model reflects a shift toward mutual contribution, where humans and machines work together to solve problems.

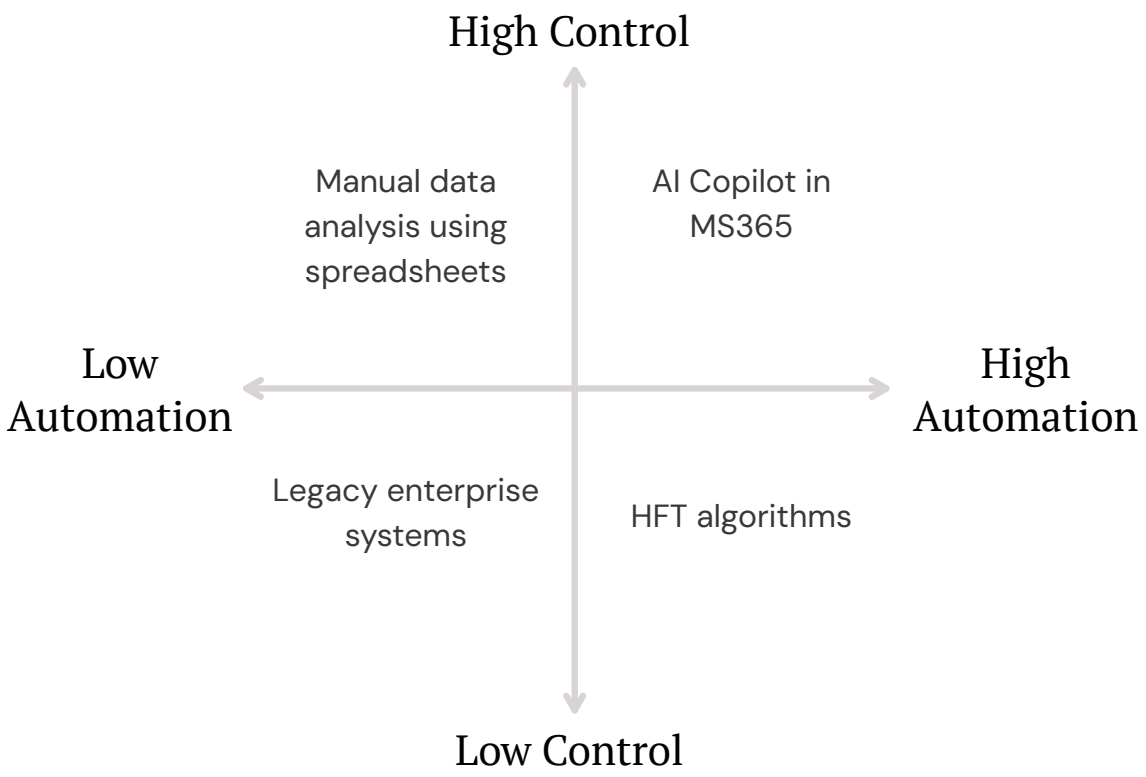
Example - Github Copilot

Trusted Advisor

The machine plays a proactive and supportive role, offering suggestions that help users make better decisions. Trust is key in this relationship, as the machine must demonstrate accuracy, transparency, and ethical responsibility.

Example - IBM Watsonx

AI Automation and Control



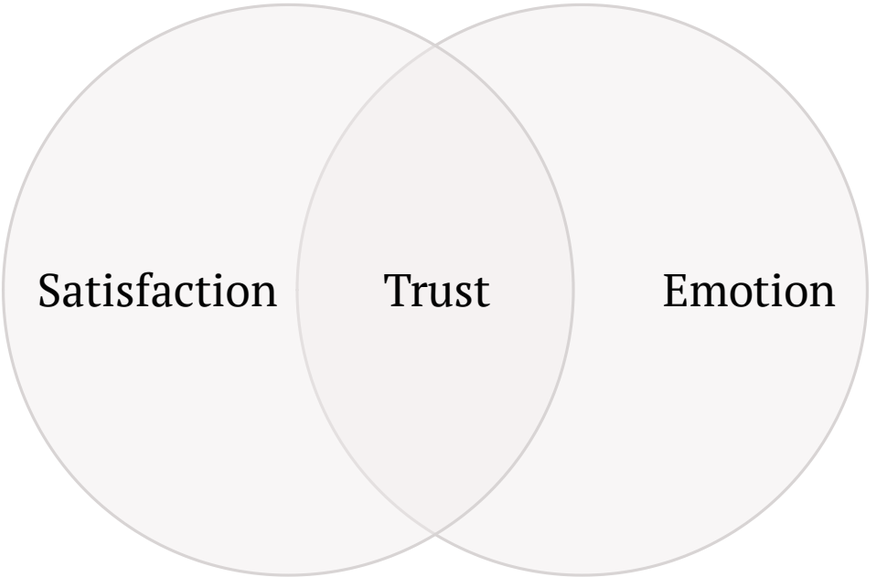
Low Automation / High Control: The user maintains full oversight, but tasks are time-consuming and repetitive.

High Automation / High Control: Good balance—AI automates repetitive tasks while users guide and override decisions as needed.

Low Automation / Low Control: Inefficient and hard to manage—neither automation nor meaningful user input is available.

High Automation / Low Control: Fast and efficient but risky. Systems make decisions without human input, which can lead to bias, misinformation, or user mistrust.

Satisfaction, Emotion, and Trust



- **Apple Case (Satisfaction + Positive Emotion = Trust):**
  A smooth, supportive service experience leaves the customer feeling valued, building lasting trust in the brand.
- **Airline Case (Low Satisfaction + Negative Emotion = No Trust):**
  Poor communication and lack of support during a flight cancellation lead to frustration and a total loss of trust.
- **Hospital Case (High Satisfaction + Negative Emotion = Weak Trust):**
  Even with successful treatment, cold and impersonal care causes emotional discomfort, weakening trust in the provider.

User Satisfaction

Satisfaction occurs when a product or service meets or exceeds consumer expectations, leading to a sense of fulfillment.

Emotional AI

Emotions are internal states that influence attention and behavior, ranging from positive feelings like curiosity to negative ones like fear.

Building Trust

Trust is built when someone is seen as capable, ethical, and fair, creating confidence in their actions and decisions.

Cognition as a Service

What is cognition?

The mental process of acquiring, understanding, and using knowledge—includes perception, attention, memory, reasoning, and decision-making.

A cloud-based offering that provides AI-powered cognitive capabilities—like language understanding, image recognition, decision-making, and sentiment analysis—so businesses can integrate human-like intelligence into their applications without building complex AI systems from scratch.

Multi-Modal Interaction

Multimodal interaction refers to user interfaces that allow input and output through multiple modes, such as speech, text, gestures, touch, and visual interfaces.

Voice AI

Voice AI enables machines to understand, interpret, and respond to human speech, transforming how we interact with technology through natural, hands-free communication.

- **Amazon Alexa & Google Assistant:** Revolutionized smart home control and voice-based search.
- **OpenAI's ChatGPT with voice:** Enabled real-time, conversational AI with emotional tone and contextual memory. (Anthropomorphism)

AI Agents and LLMs

1

AI Agents

AI agents are autonomous software entities that act independently on behalf of users to achieve specific goals. They perceive their environment and take actions without direct human intervention. Examples include smart assistants and recommendation systems.

Examples: OpenAI's Operator: Designed to autonomously perform tasks through web browser interactions. These tasks include filling out forms, placing online orders, scheduling appointments, and other repetitive browser-based activities.

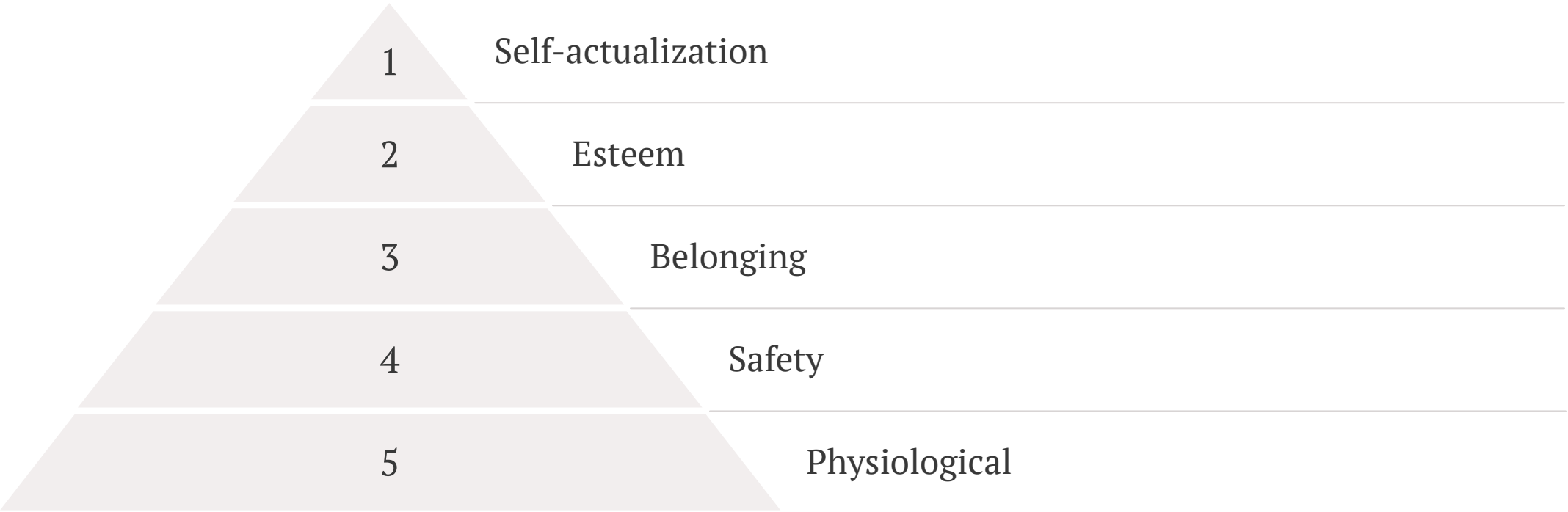
2

LLMs

LLMs are deep learning models trained on vast amounts of text data, enabling them to understand and generate human-like text. They are the driving force behind conversational AI and various content generation applications.

Examples: OpenAI's GPT 4 models, Google's Gemini

Maslow's Heirarchy Needs



Individuals generally address lower-level needs before progressing to higher-level ones. However, it's common for people to pursue multiple levels simultaneously. For instance, someone might seek both safety (a lower-level need) and esteem (a higher-level need) at the same time. Additionally, lower-level needs tend to be more universal, affecting a broader segment of the population.

Human-Centered AI

The Human-Centered AI approach to designing artificial intelligence systems puts **human needs, capabilities, and experiences at the forefront**. It emphasizes that AI should be developed and deployed in a way that enhances human well-being, improves user experience, and empowers individuals.

Amplify

To boost human capabilities by providing AI tools that extend what people can already do, making them more efficient and effective.

Augment

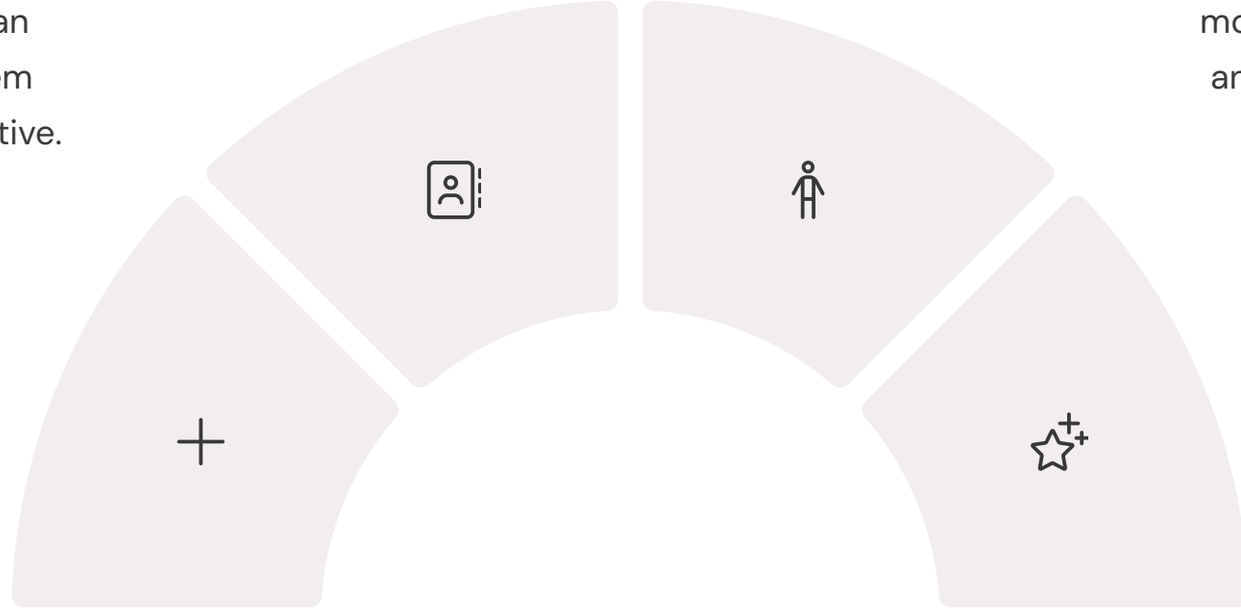
To supplement human skills by using AI to fill in gaps, handle complex tasks, or provide insights that enhance human decision-making.

Empower

To give users greater control through AI systems that are transparent, understandable, and allow individuals to direct and shape their interactions.

Enhance

To improve the overall human experience and quality of life by leveraging AI to create more intuitive, personalized, and beneficial interactions and outcomes.



Ethics in AI

1

Fairness, Bias, and Accountability

- AI systems must be designed and audited to minimize bias and ensure that decisions do not unfairly discriminate against any group.

Mitigating biases helps prevent harmful societal outcomes (e.g., discrimination in hiring or lending).

2

Privacy and Data Protection

- Protecting users' personal data is paramount; AI systems should ensure transparency over data usage and incorporate robust data protection protocols. Users should be able to edit, access or delete their data anytime.

Transparent data practices and strict security measures enhance trust among users and stakeholders.

3

Human Oversight and Ethical Governance

- AI development and deployment should incorporate continuous human oversight and involve ethical review committees to steer the technology in socially beneficial directions.

Active human supervision helps identify and mitigate potential risks from AI, such as unintentional harm or deviations from ethical norms.

Explainable AI

Transparency and Trust

- Explainable AI (XAI) methods are essential for making complex models understandable to end users, thus cultivating trust in AI decisions.
- Transparent models allow for easier identification and correction of errors and biases, leading to higher safety and reliability in critical applications (e.g., healthcare).

Balancing Performance with Explainability

- There is often a trade-off between the accuracy of highly complex “black-box” models and the interpretability of simpler “glass-box” models; a balanced approach is necessary.

Privacy Enhancing Technologies

- **Homomorphic encryption** lets you perform mathematical operations directly on encrypted data, producing results that, once decrypted, match the outcome of operating on the original plaintext.
- **Differential privacy** protects individual records by injecting randomized “noise” into query results, ensuring that no single person’s data can be reverse-engineered from aggregate statistics.